RAHUL DAHIYA

Data & BI Analyst | Python, SQL, Power BI | GenAI Applications

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SUMMARY

Data and BI analyst with 2+ years across supply chain and transportation analytics. I build Python and SQL pipelines, Power BI models, and API automations over 1 lakh+ records of SAP and operational data, and at a multi-plant manufacturer these replaced manual daily reporting with automated data reporting used in management reviews. Now applying the same data foundation to GenAI systems: RAG, text-to-SQL, and tool-calling agents built with FastAPI and MongoDB.

EXPERIENCE

Business Intelligence Executive | Servotech Renewable Power System Ltd. Jun 2026 – Present

Supply chain and logistics analytics for a multi-plant manufacturer

- Replaced manual spreadsheet consolidation with a Python ETL pipeline that loads SAP billing and sales exports and operational trackers into a SQL database, standardising 1 lakh+ dispatch, billing, and freight records into one source for reporting and reconciliation.
- Own dispatch data and reporting end to end, delivering daily planned-vs-actual, same-day freeze plan, and next-day plan reports that senior management uses to track dispatch performance and follow up on held orders.
- Removed daily manual data entry from the dispatch approval process by automating it with Google Apps Script and the Sheets, Drive, and Gmail APIs, including rule-based live sync, multi-source lookups for missing vehicle, freight, and distance data, and auto-created evidence folders.
- Rebuilt the Power BI dispatch model (Power Query, DAX) on ~20,000 rows of dispatch history with live source connections, giving management one view of dispatch value, freight cost per order, transporter performance, and plan adherence.
- Standardised dispatch KPI definitions, including on-plan versus rescheduled dispatch with holiday rules and hold-reason tracking, so plant and sales teams report against the same numbers.
- Led data reconciliation for a full finished-goods store audit, matching physical counts across racks, floors, and outdoor storage to system stock and delivering an item-wise variance report signed off by store management.
- Leading the design of an in-house Transport Management System and supporting management in evaluating a commercial TMS vendor.

Business Analyst | BT Data and Surveying India Pvt. Ltd. Oct 2024 – May 2026

Transportation analytics and client reporting

- Cut manual effort in client reporting by automating preprocessing and validation of large-scale traffic count data with Python and SQL.
- Built ETL pipelines that cleaned, transformed, and standardised high-volume transportation datasets, giving analysts consistent inputs across client projects.
- Improved data quality and reporting accuracy by designing rule-based anomaly detection checks that flagged bad records before delivery.
- Developed Power BI dashboards for traffic trends, KPI monitoring, and peak-hour analysis used in client deliverables.
- Built trend forecasting models on traffic patterns to support transportation planning studies.

Software Engineer Intern | Tata Elxsi, Bengaluru Feb 2024 – Jun 2024

Data engineering and analytics

- Gave operations teams live data visibility by designing and deploying REST APIs and real-time dashboards.
- Reduced query execution time under high load by optimising SQL queries and backend workflows.
- Contributed to containerised data processing pipelines using Docker and Kubernetes.

PROJECTS

Plant Logistics Automation Suite Ongoing

Google Apps Script, Google Sheets, HTML web app, Power BI, SAP exports

- Designing an in-house TMS for a plant handling 10–30 dispatches a day, covering the flow from SAP sales order and stock and payment checks through vehicle placement, invoicing, LR, and e-way bill.
- Built a transporter e-bidding module that sends trip details to selected transporters by email and one-click WhatsApp, collects quotes through a web form, and shortlists the lowest bid, replacing manual freight requests over WhatsApp.
- Engineered around SAP GUI-only access (no API or OData) using structured exports and a central transporter master, on zero-cost Google Workspace tooling.

Enterprise Knowledge Assistant (production RAG with evaluation) GitHub

Python, OpenAI / Claude API, MongoDB Atlas Vector Search, BM25, FastAPI, Docker, GitHub Actions

- Built a cited-answer assistant over company policies and SOPs using hybrid retrieval (vector + BM25 keyword search, Reciprocal Rank Fusion, re-ranking) with metadata filters pushed into MongoDB Atlas $vectorSearch.
- Designed a 30-question evaluation suite covering hit rate, MRR, answer accuracy, faithfulness and refusal of unanswerable questions; the offline baseline scores 1.00 hit@5, 0.96 answer accuracy and 0% hallucination, enforced as a CI gate.
- Served through a streaming FastAPI API (SSE, file upload, API-key auth), containerised with Docker; 46 automated tests.

Text-to-SQL Analytics Agent GitHub

Python, PostgreSQL, SQLite, MongoDB, sqlglot, LLM tool-calling, FastAPI, Streamlit

- Built an agent that turns plain-English business questions into SQL, results, charts and a summary, with follow-up questions ("now split that by region") resolved from MongoDB conversation memory.
- Made it safe for real databases: sqlglot syntax-tree checks (SELECT-only, table allow-list, enforced LIMIT), read-only connections, statement timeouts, and LLM self-correction from database errors.
- Defined metrics once in a YAML semantic layer that also blocks double-counting joins; reached 24/24 execution accuracy against hand-written gold SQL on both SQLite and PostgreSQL; 71 tests.

AI Integration Agent (MCP server and REST API orchestration) GitHub

Python, MCP Python SDK, FastAPI, httpx, OAuth2, webhooks, SQLite, Docker

- Built a support copilot that resolves tickets end to end across order, helpdesk and email APIs, triggered by HMAC-signed webhooks with replay protection and de-duplication.
- Engineered a resilient API layer: API-key, OAuth2 client-credentials and bearer auth, token-bucket rate limiting, exponential backoff honouring Retry-After, and idempotency keys so a lost response never causes a double refund.
- Exposed the integrations as an MCP server (9 tools, resources, prompt) sharing one tool registry with customer-scoped permissions, human approval for high-risk actions and a PII-masked audit log; 59 tests.

Data & AI Lab (supply chain and operations analytics projects) GitHub

Python, pandas, NumPy, DuckDB, SQL, PyYAML, pytest, GitHub Actions

- Build each project over three days the way a team works: plan issue, pull request, code review, issues for findings, fixes with tests, CI, then merge.
- **Rate Card History Loader:** loads an at-least-once CDC stream of carrier rate changes into an SCD Type 2 table in DuckDB with an LSN watermark, applied-event ledger and quarantine, then re-rates invoices at the rate valid on the ship date; a replayed batch changed nothing and all 21 planted billing errors were found with no false flags; 89 tests.
- **Delivery Exception Triage Agent:** tool-calling agent for delivery exception tickets with typed tool-argument checks, policy guardrails run before every refund, reship or redelivery, and fallback escalation for blocked actions, planner failures and step-budget overruns; resolved all 300 sample tickets correctly (from 97.7%) and 17 of 20 held-out messages; 129 tests.
- **Dispatch Metrics Layer:** semantic layer that defines dispatch KPIs (on-time rate, OTIF, cost per km) once in YAML and compiles any grouping, filter and time grain into one parameterised DuckDB query, with checks for duplicate fact rows and join fan-out; a reconcile report showed on-time over dispatched instead of delivered shipments is 1.6 points lower overall and 23.3 points lower in the latest week; 93 tests.

SKILLS

Programming & Data: Python (Pandas, NumPy), SQL, ETL pipelines, CDC and SCD Type 2 loading, data modelling, data contracts and data quality checks, EDA, Google Apps Script, Node.js

Databases: PostgreSQL, DuckDB, MongoDB (Atlas Vector Search), Snowflake

GenAI & ML: RAG, LLM tool-calling agents and guardrails, MCP, prompt engineering, LLM evaluation, retrieval evaluation (BM25, recall@k, MRR), LangChain, FAISS, OpenAI and Claude APIs, scikit-learn, forecasting (including intermittent demand)

APIs & Deployment: FastAPI, Flask, REST API integration, Streamlit, Docker, Kubernetes, AWS, Git

BI & Reporting: Power BI, Power Query, DAX, Advanced Excel, Google Sheets, KPI design, semantic metrics layers

Domain: SAP (SD/MM data extraction), dispatch planning, transport management, freight cost analysis, inventory audit; LLM training data work (rubric writing, response rating, safety evals)

EDUCATION

M.Sc. Computer Science (Data Analytics) 2022 – 2024

Kerala University of Digital Sciences, Innovation and Technology, Thiruvananthapuram